

The Effect of an Equatorial Continent on the Tropical Rain Belt. Part II: Summer Monsoons

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ABSTRACT: The TRACMIP (Tropical Rain Belts with an Annual Cycle and Continent Model Intercomparison Project) ensemble includes slab-ocean aquaplanet control simulations and experiments with a highly idealized narrow tropical continent (0°–45°W, 30°S–30°N). We compare the two setups to contrast the characteristics of oceanic and continental rainbands and investigate monsoon development in GCMs with CMIP5-class dynamics and physics. Over land, the rainy season occurs close to the time of maximum insolation. Other than in its timing, the continental rainband remains in an ITCZ-like regime akin to deep-tropical monsoons, with a smooth latitudinal transition, a poleward reach only slightly farther than that of the oceanic ITCZ (about 10°), and a constant width throughout the year. This confinement of the monsoon to the deep tropics is the result of a tight coupling between regional rainfall and circulation anomalies: ventilation of the lower troposphere by the anomalous meridional circulation is the main limiting mechanism, while ventilation by the mean westerly jet aloft is secondary. Comparison of two subsets of TRACMIP simulations indicates that a low heat capacity determines, to a first degree, both the timing and the strength of the regional solstitial circulation; this lends support to the choice of idealizing land as a thin slab ocean in much theoretical literature on monsoon dynamics. Yet, the timing and strength of the monsoon are modulated by the treatment of evaporation over land, especially when moisture and radiation can interact. This points to the need for a fuller exploration of land characteristics in the hierarchical modeling of the tropical rainbands.

KEYWORDS: Atmosphere-land interaction; Atmospheric circulation; Large-scale motions; Monsoons; Ensembles; Idealized models

1. Introduction

The last 20 years have seen much progress toward a theory of monsoon circulations (Geen et al. 2020). It has become apparent that individual regional monsoons should not be regarded as the product of local land–sea contrast (Gadgil 2003), but rather as elements of a coherent global monsoon (Wang and Ding 2008), integral parts of the planetary Hadley circulation and of the intertropical convergence zone (ITCZ). This recognition has led to theories of monsoons that rely only on zonal mean dynamics (Bordoni and Schneider 2008; Schneider et al. 2014). Nevertheless, for the zonal circulation to achieve its solstitial, approximately angular momentum–conserving regime the surface boundary must have low heat capacity (Geen et al. 2019). On Earth, this means that continents are necessary, that the oceanic ITCZ would not behave as the observed zonal mean rainband behaves, and that, instead, the regional monsoons shape the seasonality of the zonal mean circulation. These considerations imply that Earth’s zonal asymmetries and localized monsoons are essential to the zonal mean circulation (Dima et al. 2005; Shaw et al. 2015). Recently, Geen et al. (2019) have argued that there exist two classes of monsoon circulations, one that behaves more like a canonical ITCZ, with smooth seasonal transitions and weaker overturning circulation, and one that is characterized by abrupt onset and an angular momentum–conserving

cross-equatorial cell. In the first class are those monsoons that are confined to about 10° on either side of the equator (such as the West African and the Australian monsoon) and in the second class are those monsoons that are centered at more subtropical locations (e.g., the Indian monsoon). But what determines the location of monsoon rainfall? We still lack a theory of the tropical rainbands that is complete enough to predict this from first principles (Biasutti et al. 2018; Hill 2019).

The Tropical Rain Belts with an Annual Cycle and Continent Model Intercomparison Project (TRACMIP; Voigt et al. 2016) was implemented to address the relationship between monsoons and the ITCZ in a set of climate models with CMIP5-class dynamics and physical parameterizations. The experimental design *assumes* that the presence of a tropical continent will generate a monsoon: the control setup is a slab-ocean aquaplanet while the monsoon setup includes an idealized rectangular continent straddling the equator. In a companion paper (Biasutti et al. 2021) we focused on how the regional monsoon circulation affected the annual mean state of the ITCZ. In this paper, we focus on the monsoon circulation itself.

The first task of this study is to compare the simulated continental and oceanic rainbands to each other and to measures of the monsoon and ITCZ “regimes.” Does the continental rainband in TRACMIP show an enhanced poleward movement (Geen et al. 2019) or extent (Gadgil 2003)? Does it transition between the dry and rainy seasons with the rapidity of a monsoon (Bordoni and Schneider 2008) or the smoothness of

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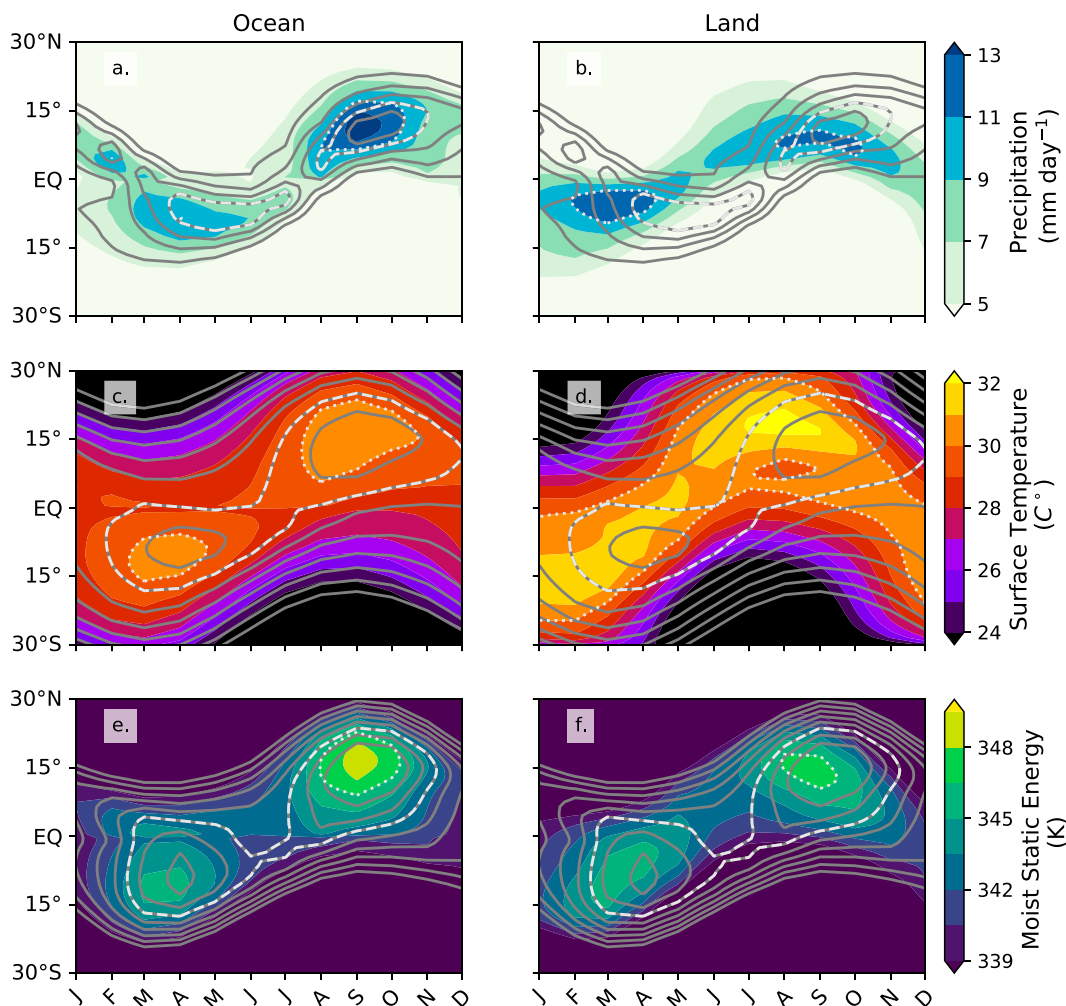


FIG. 1. Latitude–month Hovmöller diagram of the climatological (left) ocean and (right) continent sector mean of the LandControl (a),(b) precipitation, (c),(d) surface temperature, and (e),(f) low-level MSE. The AquaControl fields are repeated as contours superimposed on the LandControl shading for reference. In all panels, the same contour level is dashed in the AquaControl and dotted in the LandControl to facilitate the comparison of the timing and intensity of the maxima in each field (refer to the color bar to identify the values corresponding to the highlighted contours).

an ITCZ (Geen et al. 2019)? How sensitive is the spatial extent of the monsoon to commonly used definitions based on wind (Ramage 1971), or rainfall (Webster et al. 1998; Wang and Ding 2008)?

As we will show (e.g., in Figs. 1a,b), the continental rain-band remains confined to the deep tropics (even though the continent itself extends into subtropical latitudes) and evolves in an “ITCZ-like” regime reminiscent of the West African monsoon. In today’s Africa, the limited reach of the monsoon is ascribed primarily to the presence of the desert to the north, which both reduces the energy input absorbed by the atmospheric column (Charney 1975; Chou and Neelin 2003) and is the source of low moist static energy (MSE) advected by the regional circulation (Hill et al. 2017). The TRACMIP setup, however, does not include deserts, and thus the confinement of the monsoon has a different origin. The second task of this study is to determine what that is.

It has been argued that the poleward reach of the tropical rainfall is limited by influxes of low MSE [the literature refers to this process as “ventilation,” expanding on the original meaning of the term in Chou et al. (2001)]. In Earth-like planets, the connection between rainfall and MSE is qualitatively understood in terms of two processes fundamental to tropical dynamics: the vertical mixing due to moist convection and the horizontal temperature homogenization due to gravity waves. Convective quasi-equilibrium theory (CQE; Emanuel et al. 1994) postulates, in its simplest form, that convection relaxes the full tropospheric column to a neutrally stable profile. The strongest convection warms the column the most and, as the warming is homogenized in the free troposphere, increases the stability of the entire tropics (Sobel and Bretherton 2000; Chou and Neelin 2004; Zhang and Fueglistaler 2020). Thermodynamics would therefore predict that maximum rainfall in the tropics coincide with maximum subcloud MSE

(Chou and Neelin 2004) and, moreover, a proportionality between the two quantities (Hurley and Boos 2013; Smyth and Ming 2021). Complications arise because of entrainment, downdrafts, and differences in relaxation times between the lower (and moister) and upper (and drier) troposphere (e.g., Arakawa and Schubert 1974; Raymond 1995; Kuang 2010; Tulich and Mapes 2010). This leads to the need to consider MSE above the boundary layer; the column integrated MSE, therefore, becomes a useful bulk diagnostic for rainfall (e.g., Chou et al. 2001; Chou and Neelin 2003; Hill et al. 2017).

For both oceanic and continental tropical rain, an influx of low MSE (what we have termed ventilation) can come from the colder midlatitudes (e.g., Chiang and Bitz 2005; Kang et al. 2008; Peterson and Boos 2020)—unless the airflow is blocked by mountains, as is the case for the Indian monsoon (e.g., Boos and Kuang 2010). For the regional monsoons, low MSE can additionally come from dry deserts (Hill et al. 2017) or cool oceans (Chou et al. 2001). Land borders are therefore a key control of ventilation. TRACMIP's design does not include different land geometries: our investigation is limited to a continent 45° wide in longitude and confined to latitudes between 30°N/S. We should not expect, a priori, that our results will apply to different continental configurations, such as subtropical continents. On the other hand, the idealized studies that have linked the spatial distribution of monsoon rainfall to land geometry (Chou et al. 2001; Maroon and Frierson 2016; Zhou and Xie 2018; Hui and Bordonì 2021) are limited to one or two models and typically idealize the atmosphere severely, either by including only deep vertical modes in the circulation or by using simplified convection, clouds, and radiation schemes. TRACMIP's full-dynamics, full-physics, multimodel framework thus provides an important complementary assessment of the mechanism of ventilation and its effect on monsoon extent.

Chou et al. (2001), using a continental geometry similar to TRACMIP's, ascribed the limited monsoon extent to the transport into the eastern domain of cool, marine air by a combination of the mean westerlies and an interactive “Rodwell–Hoskins” Rossby wave emanating from the monsoon rainfall itself. Their atmospheric model (QTCM; Neelin and Zeng 2000) only allowed for the barotropic and the first baroclinic mode of circulation, so both the mean westerlies and the anomalous circulation were features of the free troposphere. Zhou and Xie (2018), using a model with simplified physics but fully resolved vertical structure, also explained the ventilation of a simplified zonally confined continent in terms of the free tropospheric westerlies. Specifically, they claimed that westerlies bring colder temperature from the ocean over the continent and, as convection homogenizes the cooling down to the surface, they end up stabilizing the atmosphere and reducing rainfall. But conclusions from these earlier studies might depend on their severe idealizations of the atmosphere and, indeed, they seem at odds with our previous results in Biasutti et al. (2021): in TRACMIP, land influences the ocean downstream via boundary layer winds, the anomalous circulation is important, and so are moist radiative feedbacks. Therefore, we examine in detail the mechanisms of monsoon ventilation.

While the atmosphere in the TRACMIP models is simulated with full physics and full dynamics, the land surface is extremely idealized: the “continent” consists of modified slab-ocean aquaplanet grid cells with increased evaporative resistance, increased albedo, reduced heat capacity, and no ocean heat transport (as specified by q fluxes). TRACMIP was not purposefully designed to explore the role of different idealizations, but fortuitous errors of implementation allow us to gain insight on the effects of each land characteristic. We have shown in Biasutti et al. (2021) that changes in heat capacity play a predominant role in the creation of solstitial anomalies over land and even of the annual mean anomalies over the ocean. Here, we again compare simulations where the continent has either reduced or unchanged heat capacity to show how the latter affects the continental rainband, in comparison to other land characteristics.

This paper is organized as follows. In section 2 we describe in more details the model simulations and our analysis procedures. The following three sections contain the bulk of our results. First (section 3), we provide an overview of the seasonal changes in the LandControl simulations and characterize the behavior of the oceanic and continental rainbands in terms of a set of descriptive measures of the monsoon and ITCZ “regimes”. Second (section 4), we provide more detail on the spatial pattern and poleward reach of the precipitation anomalies over the summer continent and we investigate whether ventilation is achieved by free-tropospheric or boundary layer winds and by the mean or the anomalous circulation. And third (section 5), we clarify the importance of a reduced heat capacity in driving the continental anomalies. Section 6 summarizes our results, discusses them in connection to previous idealized modeling of the monsoon, and provides our outlook for future research.

2. Data and methods

a. The TRACMIP protocol

Table 1 provides a list of TRACMIP models (Voigt et al. 2016) included in this study. All of the models include clouds and water vapor–radiation interactions, except the Caltech-Gray model, which assumes a fixed emissivity in the atmosphere and contains no clouds (Bordonì and Schneider 2008). We compare AquaControl and LandControl simulations. AquaControl is an aquaplanet configuration with a slab ocean of 30-m depth, zero eccentricity, atmospheric CO₂ concentrations of 348 ppmv, and a prescribed ocean heat transport convergence that is an idealized version of the observed zonal mean and that is the only source of asymmetry in the simulations under consideration. Because of this ocean heat flux, the NH is warmer than the SH in the annual mean. LandControl includes an idealized continent 45° wide in longitude and extending in latitude from 30°N to 30°S. The idealization of land properties is accomplished by modifying ocean grid cells in the following ways: 1) the q fluxes representing ocean heat transport convergence are zeroed out in the continent region (note that a uniform compensation over the ocean ensures zero net energy flux anomaly in the global mean); 2) the

TABLE 1. List of atmospheric GCMs used in this study, along with the coupled Earth system model that each atmospheric model is a component of, if applicable. Only the models following protocol (not starred) are included in the multimodel means. All models except CaltechGray are full physics GCMs. Citations and additional details such as model resolution are listed in Voigt et al. (2016).

Atmospheric model	Component of	Protocol
CaltechGray	—	Yes
CAM3	CCSM3	Yes
CAM4	CCSM4	Yes
CAM5Nor	NorESM2	Yes
CNRM-AM5	CNRM-CM5	Yes
ECHAM6.1	MPI-ESM	Yes
MetUM-CTL*	GA6.0	No: heat capacity as in AquaControl
MetUM-ENT*	GA6.0 (modified)	No: heat capacity as in AquaControl
MIROC5 (atmospheric component)	MIROC5	Yes
MPAS (atmospheric component)	MPAS	Yes

surface albedo over the continent is increased by 0.07, corresponding to vegetated land; 3) the evaporation rate coefficient in the bulk moisture flux equation is halved, representing enhanced evaporative resistance by vegetation; and 4) the heat capacity is reduced by changing the mixed layer depth from 30 m (similar to the tropical mean mixed layer depth and appropriate to achieve an Earth-like annual cycle in the aquaplanet configuration) to 0.1 m (representative of a shallow layer of moist soil).

We focus on the models that followed protocol exactly [hereafter, the protocol models; see the correction to Voigt et al. (2016) and discussion in Biasutti et al. (2021)], but we also briefly present results from models that incorporated the first three properties of land but did not reduce the heat capacity of the continental region (hereafter, the MetUM models). We compare the MetUM to the protocol models in order to isolate the anomalies due to the choice of mixed layer depth from those due to other land characteristics. To ensure that our interpretation is correct, and that model choice is not an issue, we ensure that the intermodel scatter across the protocol models is much smaller than the difference between the protocol models and the MetUM models.

b. Rainband diagnostics

We refer to zonal-mean quantities as “rainbands,” with the understanding that the zonal average is calculated from global data in the AquaControl simulations and over just the continent (0°–45°W) in the LandControl simulations. Land and ocean climatologies differ substantially in how fast either one responds to the external forcing coming from seasonally varying insolation. Therefore, LandControl – AquaControl differences emphasize the changes in the timing of the rainy season at any given latitude. If, instead, we compare the rainbands in their respective rainy seasons, we emphasize differences in structure and behavior, independent of timing.

We characterize the seasonal monsoon by either rainfall or wind and the year-round rainbands in terms of their spatial extent, position, rapidity of their meridional displacement, and characteristics of their ascent (vertical profiles, frequency, and intensity). Specifically, we define the monsoon regions and describe the rainbands according to the metrics summarized below.

Monsoon rainfall: Following Wang and Ding (2008), we define monsoon regimes where (i) the local summer-minus-winter precipitation rate exceeds 2 mm day^{-1} and (ii) the local summer precipitation exceeds 55% of the annual total. The first criterion distinguishes the monsoon climate from more arid climate regimes. The second ensures that precipitation is concentrated during local summer, thereby distinguishing the monsoon climate from equatorial perennial rainfall regimes. We define summer differently in the case of LandControl and AquaControl. For LandControl we take local summer to denote May through September for the NH and November through March for the SH. AquaControl seasons are shifted by three months (i.e., NH summer goes from August to December and SH summer from February through June).

Wind reversal: We identify regions of wind reversal as those regions where the maximum difference in wind direction for any pair of months is larger than 90° , for nonnegligible wind speed (the exact value of the threshold is unimportant).

Rainband position: We calculate the position of the rainband as the centroid of precipitation following the definitions of Adam et al. (2016) and Voigt et al. (2014),¹ or as the latitude of maximum rainfall.

Rainband migration speed: We take the time derivative of the 5-day running-mean smoothed daily values of the rainband position to calculate the meridional translation speed of the rainbands (Geen et al. 2019).

Rainband width: Following Byrne and Schneider (2016), we define the width of the rainbands as the meridional distance where net precipitation [precipitation minus evaporation ($P - E$)] is positive.

Rain characteristics: We diagnose changes in rainfall characteristics in terms of frequency of rainy days (rain

¹ The Adam et al. (2016) definition calculates the precipitation- and area-weighted mean of latitude between 30°N and °S; the Voigt et al. (2014) definition calculates the latitude at which the area-integrated precipitation (within the same tropical band) that falls to its north equals the area-integrated precipitation that falls to its south. The former definition is more weighted toward rainfall away from the equator and indicates a smaller seasonal excursion than the latter definition.

accumulation larger than 1 mm day^{-1}) and simple daily rain intensity (rain intensity on rainy days in mm day^{-1}).

We use climatologies based on the last 20 years of monthly data or, when daily data are necessary, on 10 years of simulations.

c. Other diagnostics

We link the position of the rainbands to simple diagnostics of the horizontal gradients in the low-level atmosphere. Specifically, we calculate the latitude of the zonal or sector mean of the following:

The intertropical front: the minimum in sea level pressure.

This is equivalent to the locus of surface mass convergence, and it is expected to be tightly related to boundary layer moisture convergence in the absence of strong moisture gradients.

The 925-hPa MSE maximum: a measure of the subcloud layer MSE maximum. From a purely thermodynamic perspective, this quantity should coincide with maximum rainfall (see the introduction). Dynamic considerations, instead, require that maximum surface MSE limit the poleward extent of the overturning cell, so that maximum vertical motion and, thus, rainfall remain on the equatorward flank [see [Privé and Plumb \(2007\)](#) for a derivation based on axisymmetric theory and [Singh \(2019\)](#) for an extension].

The surface temperature maximum: the connection between SST and rainfall is not direct, but instead it is mediated by sea level pressure ([Lindzen and Nigam 1987](#); [Back and Bretherton 2009](#)) and MSE ([Emanuel et al. 1994](#); [Hurley and Boos 2013](#)). Yet it remains a commonly used and useful diagnostic ([Biasutti et al. 2021](#); [Wei and Bordoni 2018](#)), and we report it here.

We use climatologies based on the last 20 years of monthly data.

3. Monsoon and ITCZ regimes: Diagnostics of oceanic and continental rainbands

[Figure 1](#) shows the month–latitude Hovmöller diagrams of climatological fields that have been zonally averaged over the oceanic and continental sectors in the LandControl simulations; contours of the AquaControl climatology (zonally averaged) are superimposed on the LandControl sector averages in order to help the comparison. Besides the rainbands ([Figs. 1a,b](#)), we show the seasonal evolution of surface temperature ([Figs. 1c,d](#)), and low-level MSE ([Figs. 1e,f](#)). The climatology of the LandControl simulation averaged over the ocean sector (left panels, [Figs. 1a,c,e](#)) is similar to the zonal mean of AquaControl, with only small differences in the timing and intensity of peak anomalies in all fields. This similarity is consistent with our findings in [Biasutti et al. \(2021\)](#), in which we show that the influence of land extends only about 120° to the west of the continent, leaving most of the ocean unaffected. In what follows, we select to directly contrast the oceanic rainband in the AquaControl and the continental rainband in the LandControl.

The most obvious difference between LandControl and AquaControl, and the expected result of a reduced surface heat capacity, is that the annual cycle is phase-shifted early over land, compared to the ocean, by between 1 and 2 months in all variables (right panels, [Figs. 1b,d,f](#)). Peak values are also affected by land characteristics, but differently for different fields. Precipitation shows a small reduction in peak values, especially in the Northern Hemisphere. The surface temperature summer-to-winter seasonal excursions are of larger magnitude in LandControl than in AquaControl (as is also expected for a lower heat capacity system forced by oscillating heat fluxes). In contrast, seasonal MSE excursions remain similar across ocean and land, but MSE is overall reduced in LandControl. A lesser MSE maximum derives from the imposed reduction in local evaporation and also from transport of low MSE into the continent (see also [section 4](#)).

The impression one derives from [Fig. 1](#) is that the oceanic and continental rainbands are overall very similar, aside from their phasing within the calendar year. Following [Gee et al. \(2019\)](#), we suggest that the TRACMIP monsoon is in a deep-tropical, ITCZ-like regime, namely a regime in which the monsoon never jumps to subtropical latitudes and never develops an approximately angular momentum-conserving circulation. Yet, the behavior of the land-based rainband remains distinct from that of the oceanic ITCZ.

More detailed analysis supports this suggestion. [Figure 2](#) shows the seasonal migration of the rainband (indicated by three definitions in different shades of blue), alongside the location of maximum surface temperature (red), maximum boundary layer MSE (magenta), and minimum sea level pressure (black). In AquaControl, the maxima of surface temperature and MSE linger at their northernmost and southernmost positions and transition between the two rather quickly, more like square waves than sinusoids. The rainband moves between latitudes in a manner that is more gradual, but also more asymmetric: the shift from south to north is quicker than the reverse. Thus, over ocean, the relationship between the rainband and the position of the maximum MSE varies over the course of the seasonal march.

We have already noted that the evolution of the continental climate is shifted early. We now see that, in the NH, the timing of extrema in surface temperature, MSE, and SLP shifts more (from October to August) than that of the rainband (from October to September). In the SH, both the rainband and the surface extrema shift by the same amount, two months. Thus, the northward migration and the southward migration are now of the same duration. Moreover, while the loci of extreme temperature, MSE, and SLP are experiencing larger meridional excursion over land than over ocean, the rainband is not: it oscillates between 5°S and 10°N over both land and ocean. This causes a larger separation between the rainband and surface extrema (temperature, MSE, SLP) over land, compared to the ocean. A separation of the rainband from the maximum in MSE is expected from theories of the zonally symmetric moist circulation, especially for ITCZs located off the equator ([Privé and Plumb 2007](#)) but within the tropics ([Singh 2019](#)). Nevertheless, it is unclear why the magnitude of this displacement would be larger over land, given

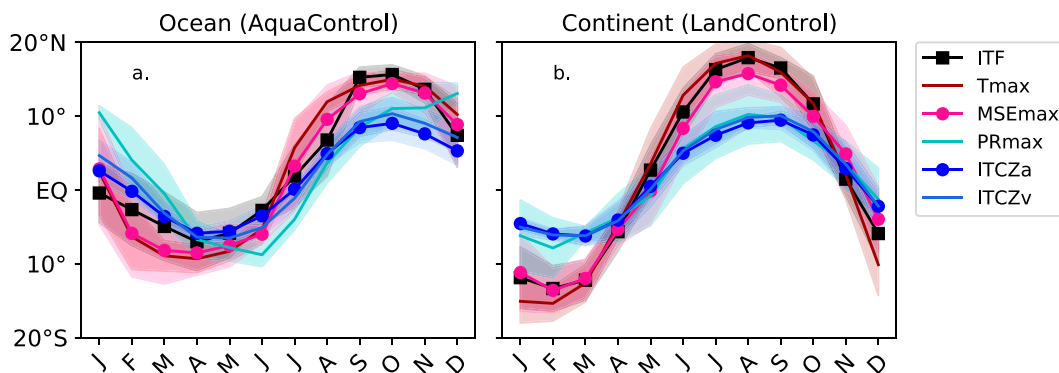


FIG. 2. Meridional positions, as a function of climatological month, of the ITCZ [ITCZa and ITCZv indicate the definitions of Adam et al. (2016) and Voigt et al. (2014), respectively; blue lines], the maximum in rainfall (light blue), the minimum in SLP [indicated as the intertropical front (ITF); black], the maximum in low-level MSE (magenta), and the maximum in surface temperature (red), for (a) the ensemble and zonal mean of the AquaControl simulations and (b) the ensemble and continent sector mean of the LandControl simulations. Shading indicates one standard deviation in the multimodel ensemble.

that the location of the rainband is similar in the two domains. A larger separation between the rainband and the ITF, compared to that seen over ocean, is a feature of real world monsoons, most famously in West Africa and Australia (Nicholson 2018; Nie et al. 2010). But the correspondence with TRACMIP is only partial: in observations the ITF pushes into dry deserts and produces dry ascent and a shallow circulation, while in the simulations, ascent remains deep between the rain centroid and the ITF, leading to rainfall. As we shall see in the next section, a zonal-mean view might be insufficient to explain the meridional extent of the TRACMIP monsoon.

The degree of similarity in the progressions of the oceanic and continental rainbands is detailed in Fig. 3. The top panels reveal that both rainbands reach similar northernmost and southernmost positions: there is less difference between the LandControl and AquaControl cases than across models of the ensemble or across two commonly used centroid definitions. The translation speeds (shown in Figs. 3c,d for one centroid definition, but robust to the choice) are also somewhat similar between ocean and land, but with some noteworthy differences. Compared to the aquaplanet, migration speeds over the continent are generally faster and less consistent with a perfect sinusoidal progression (shown as an ellipse calculated from the annual harmonic). The onset of the land monsoon (first and third quadrants) is somewhat slower than its demise (second and fourth quadrants) in opposition to the behavior of the AquaControl ITCZ and to that reported for aquaplanet monsoons in Geen et al. (2019).

Figure 4 shows the evolution of the rainband width, as defined in section 2. The two leftmost panels show latitude-month diagrams, while the right panel shows both the summer reach of the rainband in each hemisphere (vertical bars, left axis) and the maximum width of the rainband over the course of the year (markers, right axis). By either of these measures, the land-based rainband behaves in ways qualitatively similar to the ocean-based ITCZ, with the only difference that it

reaches slightly farther poleward (especially in the SH) and is slightly wider throughout the year (but not in all models).

Finally, we move past the two-dimensional view of the monsoon in Fig. 5, which shows the extent of the “global monsoon” as defined by the seasonality of rainfall and wind. The two definitions select for different regions: The rain-based monsoon region is nearly completely confined to the continent, extends to the subtropics, and is more extensive in the SH (where rainfall is concentrated in a shorter rainy season). The wind-based monsoon is elongated over the ocean, meridionally confined to the deep tropics, and is more extensive in the NH (where the circulation is stronger). The narrow extent of the wind-based definition is reminiscent of the African case. Not so the rain-based definition, which selects for subtropical areas that, in observations, are deserts (evaporation from a permanently moist surface in TRACMIP causes the discrepancy, as can be surmised from the $P - E$ pattern). Nevertheless, when we take the sector or zonal averages (right panels), both definitions are consistent with each other and with the $P - E$ metric in selecting for a slightly broader meridional span of the LandControl rainband, compared to the AquaControl.

In summary, the above analysis shows that the TRACMIP monsoons is a deep-tropical monsoon in an ITCZ-like regime, with some similarity to the West African monsoon. First, the width of the TRACMIP rainband is similar over land and ocean and close to constant throughout the year. Second, the rain’s northernmost reach is similar in the two domains. Third, areas of positive $P - E$ progress smoothly from one hemisphere to the other. Again, this behavior agrees with observations in the African sector: the maximum in rainfall jumps from the coastal ocean to the interior at the beginning of summer (Sultan and Janicot 2003), but the zonally averaged rainfall band progresses quite smoothly. Moreover, the transition over Africa is faster in its retreat than in its advance (Biasutti 2019), consistent with the behavior seen in the TRACMIP LandControl.

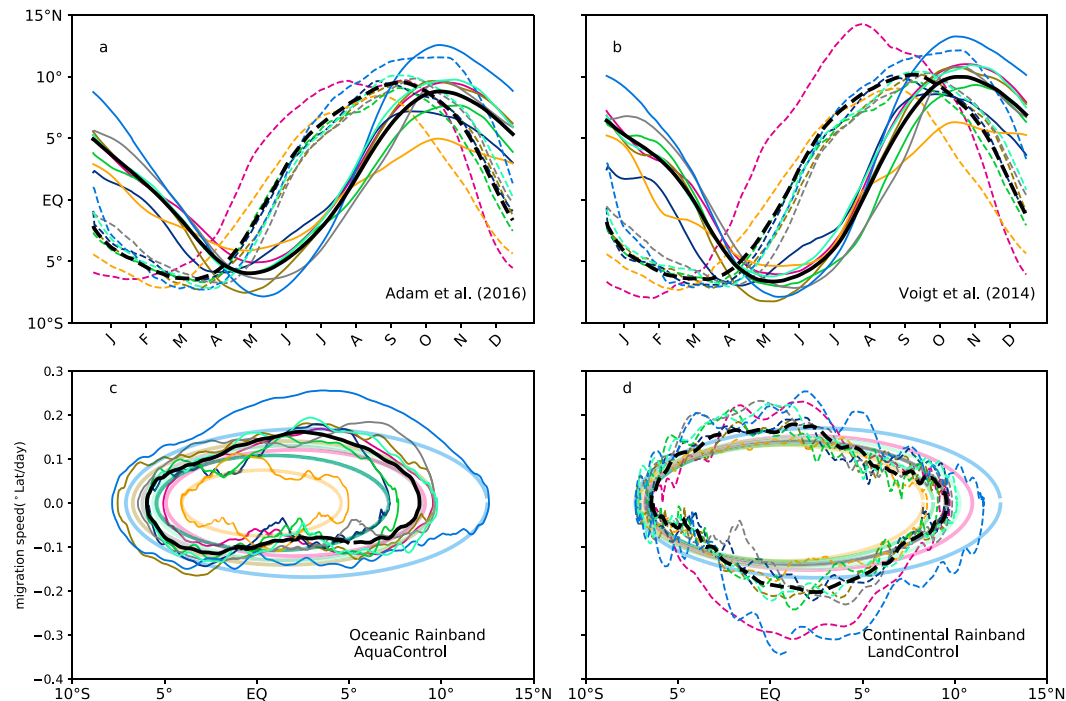


FIG. 3. (top) Position of the ITCZ in the AquaControl (solid lines) and LandControl (dashed lines) according to the definition of (a) Adam et al. (2016) and (b) Voigt et al. (2014) as calculated from smoothed daily data. (bottom) Speed of meridional migration of the ITCZ [according to the Adam et al. (2016) definition] in (c) AquaControl (solid lines) and (d) LandControl (dashed lines). The perfect ellipses in (c) and (d) are obtained by fitting the ITCZ position with two Fourier components (time mean plus a single annual sinusoid) and plotting the corresponding position and velocity. Migration speeds outside the ellipses correspond to rapid transitions. Individual models are color coded (see Fig. 4 for a legend); the multimodel mean is given by the thick black lines.

4. The poleward extent of the summer monsoon: Mechanisms of ventilation

A map view of the LandControl – AquaControl seasonal anomalies provides clues to the processes that determine the extent of the TRACMIP monsoon and indicates that zonal asymmetries are important. (The extent to which this conclusion depends on the narrow longitudinal extent of the continent is discussed later.) Figure 6 shows the surface temperature (shaded) and precipitation (contour) anomalies for the four standard seasons; the AquaControl rainband is also shown for reference. Throughout the year, temperature and rainfall anomalies over land are consistent (in sign and strength) with the accelerated response of the continent to insolation (compared to the ocean) and with the tendency for rainfall to follow the net energy input into the atmosphere. This translates to small anomalies during equinox seasons (comparable to the annual mean anomalies; Biasutti et al. 2021) and much larger anomalies during the solstice seasons. Anomalies in both temperature and rainfall are positive in the summer hemisphere and negative in the winter hemisphere.

The wintertime cold anomalies are the largest, due to the reinforcing effects of enhanced resistance to evaporation and reduced energy input, further amplified by moist-radiative feedbacks and by the divergent surface circulation (Biasutti

et al. 2021). Summertime and wintertime anomalies in rainfall are more comparable in their peak positive and negative values, but they differ greatly in shape. The wintertime dry anomalies are centered at the latitude of the AquaControl ITCZ and are roughly zonally oriented (both foregoing consequences, to some degree, of no negative rainfall). The summertime wet anomalies extend poleward from the latitude of the AquaControl ITCZ and are characterized by a triangular pattern: they are narrow in the western part of the continent and broad in the eastern part, where they reach the coastlines at 30°N/S. A similar pattern of summertime rainfall anomalies has been interpreted (Chou et al. 2001; Zhou and Xie 2018) as the effect of ventilation, primarily by the mean free tropospheric westerlies. We find that ventilation happens by different mechanisms in TRACMIP.

Figure 7 shows fields relevant to ventilation in the two summer hemispheres: JJA above the equator and DJF below the equator. The top and bottom panels describe processes in the free troposphere and in the boundary layer, respectively. Figure 7a shows temperature anomalies at 300 hPa (shaded), geopotential anomalies at 700 hPa (contours), and the full LandControl wind at 700 hPa (vectors). The mean westerlies are weak over the subtropical portion of the continent and the temperature anomalies do not resemble what we would expect from westerly advection: instead of decaying inland,

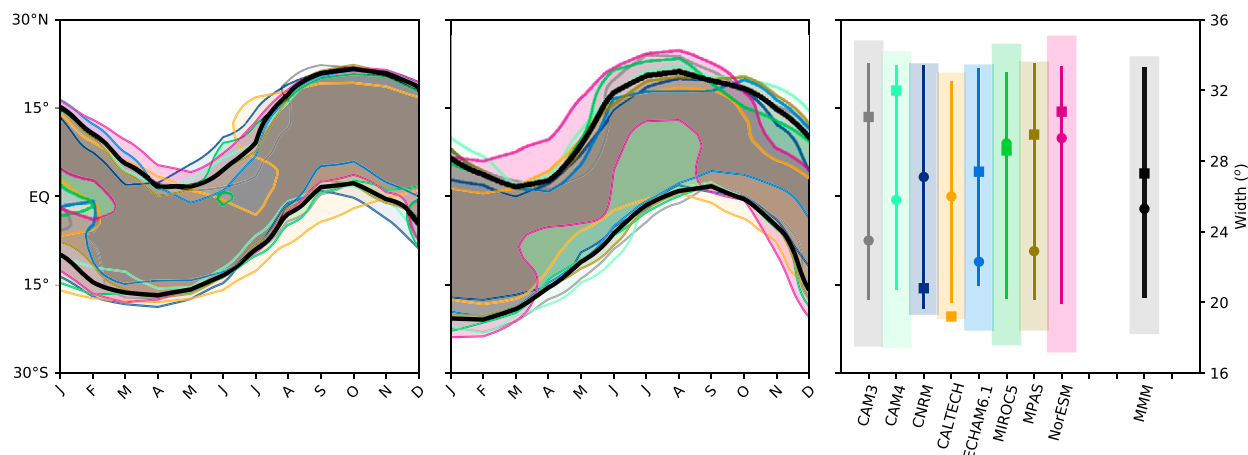


FIG. 4. Width of the ITCZ as indicated by the zero contour of the $P - E$ field. (left) The zonal mean of the AquaControl simulations, (center) the sector mean of the LandControl simulations, and (right) the meridional span of the ITCZ in its seasonal march (the southernmost reach of the southern zero $P - E$ contour and the northernmost reach of the northern zero $P - E$ contour); broad bars for the LandControl case and thin lines for the AquaControl case; y axis on the left, corresponding to that on the other panels. The right y axis measures the maximum width of the ITCZ (distance between the southern and northern zero $P - E$ contours) during the course of the year for the AquaControl zonal mean rainband (dots) and the LandControl sector mean continental rainband (squares). Individual models are color coded as indicated in the right panel; the multimodel mean is given by the thick black lines.

they are strongest in the western part of the continent and they are warm in the summer hemisphere subtropics, opposite to what is necessary for ventilation (Zhou and Xie 2018). Upper-level temperatures are cold everywhere else and show the Gill-like signature (Gill 1980) of the negative rainfall anomalies in the oceanic cold tongue. It is possible that these cold temperatures are homogenized downward by convection and modulate rainfall and surface temperature in the core monsoon region. Nevertheless, they do not appear to be preventing rainfall in the western portion of the subtropical continent.

Figure 7b shows fields relevant to low-level processes (anomalies in precipitable water, boundary layer geopotential, and wind) and suggests a predominant role for such processes in limiting the monsoon in the western portion of the continent and enhancing it in the east. Note, for example, the correspondence between the slanted positive anomalies in precipitable water over the summer continent and the low-level cyclonic circulation that brings tropical moist air to the eastern continent and subtropical dry air to the western continent.

The above suggestions are confirmed by a quantitative analysis of MSE advection. Figures 8a and 8b show the total MSE advection in the boundary layer and the free troposphere (925 and 300 hPa, respectively; these levels were chosen as the most clearly representative, but results are robust to the choice) in the NH during JJA (DJF anomalies in the SH are a nearly perfect mirror image of JJA in the NH and we omit them for clarity). The pattern of anomalies is similar at both levels, but the magnitude of the anomalies is much larger in the boundary layer. We decompose the advection in its zonal and meridional terms and further decompose those as the linear combination of the advection of anomalous MSE by the mean wind and advection of mean MSE by the

anomalous wind. We obtain four terms that are plotted in Figs. 8c–j. This decomposition highlights how MSE advection is achieved differently at different levels. In the free troposphere, the mean westerlies acting on the anomalous gradient of MSE do indeed ventilate the western part of the continent, as suggested in the literature. But this effect is counteracted by the other terms, especially by the advection of the climatological MSE gradient by the anomalous meridional wind. Within the boundary layer, the dominant mechanism of ventilation is the advection of the background MSE gradient by the meridional component of the anomalous circulation. The background zonal wind is most relevant at the coastlines, where it acts to counteract the main advection pattern. The other terms are small over the subtropical continent. (We note as an aside that the anomalous negative MSE advection that extends past the continent at about 10°N is the result of the covariant term.)

The vertical profiles of the MSE transport terms (Fig. 9) confirm the description above and add some insight into the scatter across models. Higher in the troposphere, the advection into the western subtropical continent of low oceanic MSE by the mean zonal wind is compensated by the advection of the mean MSE by the anomalous meridional wind. Each term is uncertain across the ensemble, but the cancellation is not, so that the total uncertainty in the free-troposphere ventilation is low. Lower in the boundary layer, the continent is ventilated by the anomalous meridional wind acting on the background gradient in MSE between the tropics and the midlatitudes. This is the dominant term in the column MSE budget and imparts its uncertainty to the total advection term.

We conclude that, in TRACMIP, the diffusion of MSE anomalies by the free-troposphere westerlies is an active mechanism, but not the one primarily responsible for the

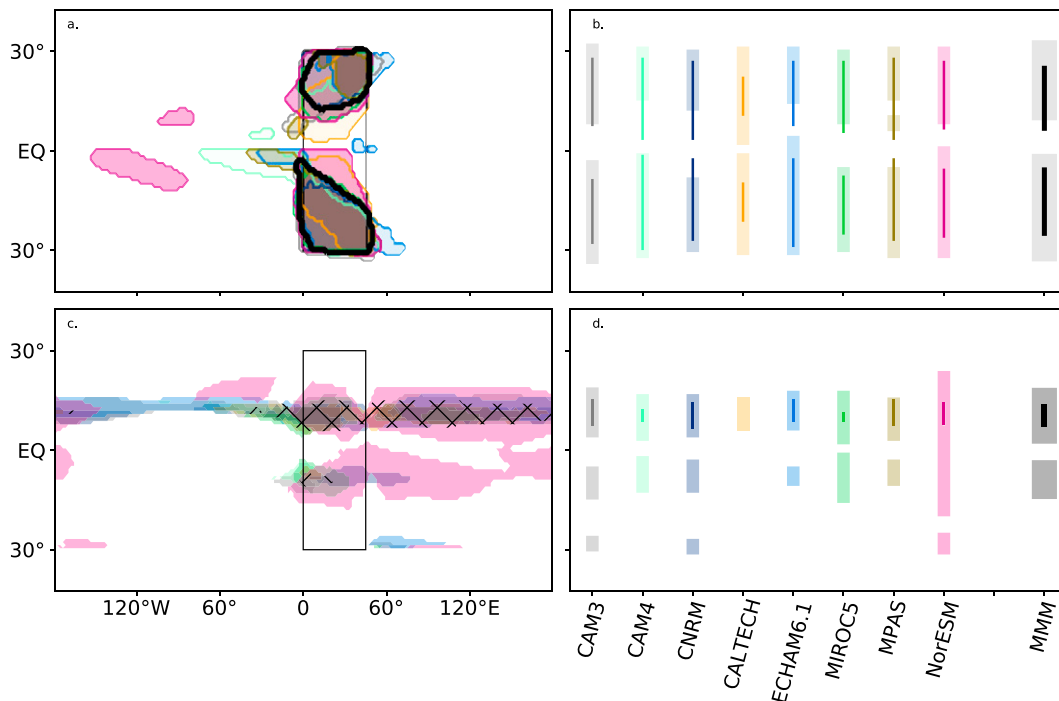


FIG. 5. (top) Spatial extent of the summer rainbands and (bottom) loci of seasonal wind reversal. (a) Map of the LandControl global monsoon, as defined by a precipitation criterion. Each model is color-coded as indicated in the right-hand panel and the multimodel mean is indicated in black. (b) Maximum meridional summer extent of the rain-defined global monsoon domain in each model (bars are for the LandControl case and thin lines are for the AquaControl case) and in the multimodel mean. (c) Map of the LandControl global monsoon, as defined by a wind reversal criterion. Each model is color-coded and the multimodel mean is hatched in black. (d) As in (b), but for the wind reversal criterion. Refer to the text for the exact definition of the rainfall and wind criteria for the global monsoon in the Land and Aqua cases.

ventilation of the subtropics. The poleward extent of the monsoon rains, in its mean and its uncertainty, is predominantly a consequence of anomalous poleward flow in the boundary layer acting on the prevailing MSE field that decreases toward the pole.

5. Land idealizations: The effect of a reduced heat capacity

From the simplest model of a uniform surface layer forced by a sinusoidal heat source, we expect that the small phase shift between insolation and surface temperature over land derives from the reduced heat capacity of continental grid points. Yet, we have seen in Figs. 1 and 3 that neither MSE nor, especially, rainfall, covaries perfectly with temperature, so that the question of the role of different land characteristics on rainfall remains somewhat open.

To identify whether land characteristics other than heat capacity contribute to the simulated LandControl – AquaControl seasonal changes, we contrast the mean anomalies across models that exactly followed the TRACMIP protocol to those across the two MetUM models, in which a reduced heat capacity for land grid points was not imposed. Figures 10a and 10b show the latitude–month Hovmöller diagrams of LandControl – AquaControl rainfall anomalies (alongside

the AquaControl rainband, for reference). The top panel shows alternating dipoles in rainfall anomalies in the protocol models, with wet anomalies preceding, and dry anomalies trailing, the AquaControl rainband. The mean state and the anomalies are close to being in quadrature, suggesting a shift in the seasonality and consistent with a much smaller annual-mean signal (Biasutti et al. 2021). The bottom panel (in which land does not have a reduced heat capacity) shows peak anomalies of similar magnitude, although the pattern is different. When idealized land retains a high heat capacity, positive equatorial anomalies persist through the year and the subtropical dry anomalies are limited to local summer, when they act to reduce the local maximum. Thus, the timing of the rainy season remains unaffected.

We check the robustness of these results by examining the rainfall anomalies in the individual protocol models and MetUM models averaged within the northern (Fig. 11a) and southern (Fig. 11b) continent. Only the protocol models show the alternating positive and negative anomalies, while the MetUM models show only dry anomalies, especially intense in correspondence of the main rainy season. We note that the CAM5-Nor model (magenta line) is an outlier among the protocol models, somewhat closer to the behavior of the MetUM models: drying associated with land characteristics besides heat capacity (evaporative resistance, albedo, and

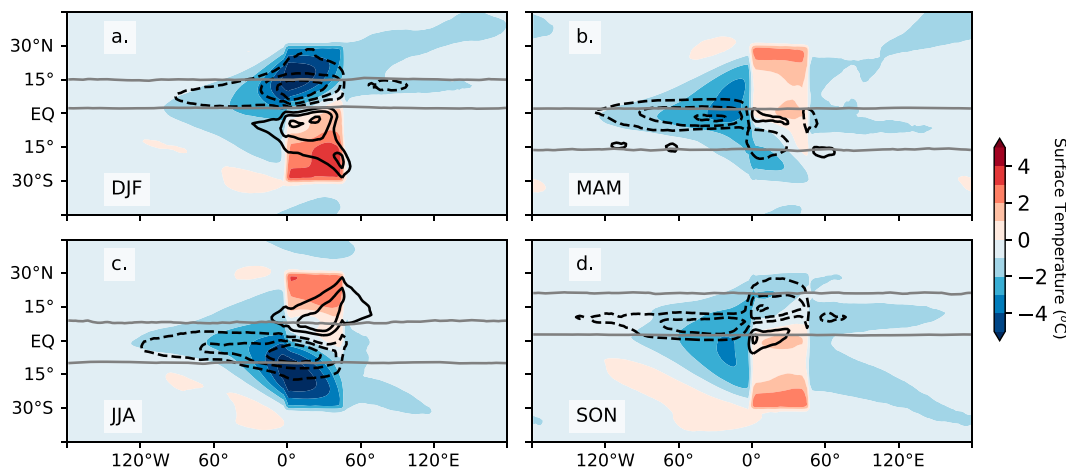


FIG. 6. Maps of the surface temperature (shaded) and precipitation (black contours, every 2 mm day⁻¹, skipping the zero contour) LandControl minus AquaControl ensemble-mean seasonal anomalies. The seasonal means are (a) DJF, (b) MAM, (c) JJA, and (d) SON. The position of the AquaControl rainband for each season (as indicated by the 6 mm day⁻¹ contour) is superimposed in gray for reference.

lack of heat transport convergence) has a more prominent role in this model. Nevertheless we will consider the ensemble mean of all protocol models and interpret mean phase shifts as due to changes in heat capacity.

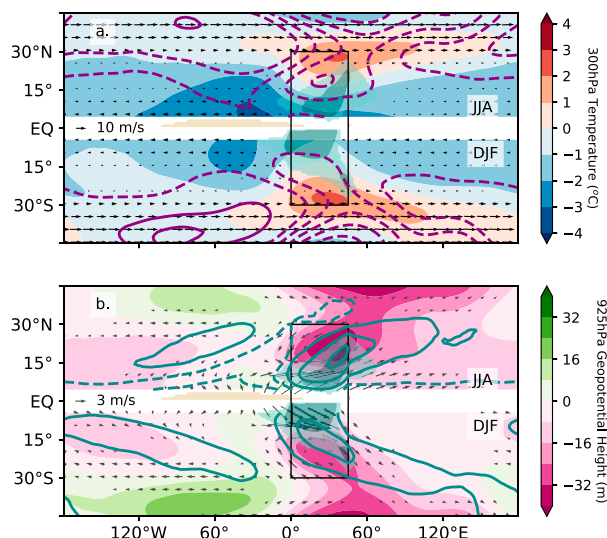


FIG. 7. (a) Summertime (JJA in the NH and DJF in the SH) ensemble mean LandControl minus AquaControl ensemble-mean anomalies in 300-hPa temperature (shaded) and in 700-hPa geopotential (contours, contour interval is 8 m; dashed lines indicate negative anomalies and the zero contour is omitted), and 700-hPa LandControl full wind field (vectors; only wind speeds greater than 3 m s⁻¹ are plotted; see reference arrow on the left). Summertime rainfall anomalies less than -2 mm day⁻¹ are shaded in brown, and those greater than 2 (4) are shaded in green (dark green). (b) As in (a), but shading indicates 925-hPa geopotential heights, contours indicate precipitable water (contour interval is 5 g kg⁻¹), and wind vectors are for the 925-hPa anomalies (only wind speeds larger than 1 m s⁻¹ are plotted; see reference arrow on the left).

Rainfall reduction in the continental subtropics occurs by different mechanisms when it is due primarily to a smaller heat capacity or primarily to a resistance to evaporation. Figures 10b and 10e show the LandControl - AquaControl changes in the frequency of rainy days in the two sets of models (protocol and MetUM); Figs. 10c and 10f show the changes in daily intensity. Peak changes in intensity are around 8 mm day⁻¹, either in positive or in negative values and in both sets of models. Peak changes in rain frequency are much larger in the case of the protocol models, and much larger for negative than for positive anomalies. This asymmetry is consistent with the more pronounced wintertime circulation changes driven by the heat capacity-induced land-sea contrast [Fig. 6; see also Biasutti et al. (2021) for a comparison with the MetUM models] and with a greater role for dynamics, as opposed to thermodynamics, in affecting the occurrence of rainy days rather than their intensity. In contrast, in the MetUM simulations, the imposed land characteristics do not create large circulation in and out of the continent and changes in rainfall are predominantly caused by thermodynamic properties and expressed as changes in intensity.

A reduced heat capacity also affects the profile of ascent in the rainband (Fig. 12). Figures 12a and 12c show the latitude/pressure zonal and sector mean of vertical velocity for the SH summer (DJF in LandControl and MAM in AquaControl in the case of the protocol models, MAM in both LandControl and AquaControl in the case of the MetUM models). Figures 12b and 12d show each model's profile in the ascent regions. For models with a reduced heat capacity over land, vertical ascent is larger in magnitude and much more top heavy over land than over ocean.² The omega profile remains unchanged

² The NorESM model is an exception, but we have not adjusted the averaging period to match its continental summer; when that is done, it too has deeper ascent over land.

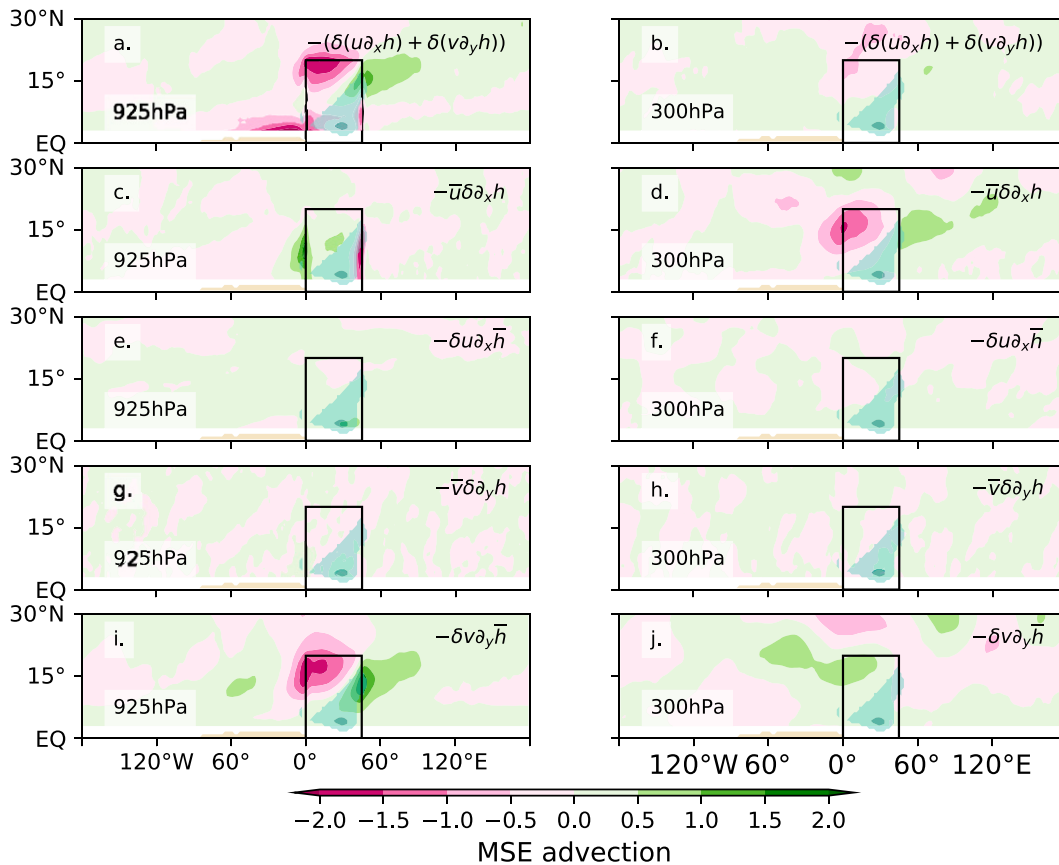


FIG. 8. Maps of NH summertime (JJA) LandControl minus AquaControl ensemble-mean anomalous MSE transport [$^{\circ}\text{C m s}^{-1} (^{\circ}\text{latitude/longitude})^{-1}$] at (left) 925 hPa and (right) 300 hPa. (a),(b) Total advection anomalies, (c),(d) advection of the anomalous MSE gradient by the basic-state zonal wind, (e),(f) advection of the basic-state MSE gradient by the anomalous zonal wind, (g),(h) advection of the anomalous MSE gradient by the basic-state meridional wind, and (i),(j) the advection of the basic-state MSE gradient by the anomalous meridional wind.

in the case when land does not have a reduced heat capacity. This change in the vertical profile of ascent only depends on the different heat capacity of the lower boundary, not on where the rainband is in its seasonal march. It follows that the presence of a low heat capacity continent will influence the responsiveness of the rainband to MSE fluxes: deeper or shallower profiles of ascent are associated with larger or smaller moist stability (Raymond et al. 2009), thus modulating the relationship between the position of the rainband and MSE transport (see, e.g., Biasutti et al. 2018). It should be noted, though, that the difference in ascent profile between the (low-heat capacity) land and the ocean is not due to the difference in the local heat capacity per se, but derives from changes in the large-scale circulation. This can be surmised from the comparison of the profile ascent in the western third of the continent with that over the eastern third of the continent, which show markedly different features (not shown). In the east, where the low-level flow is extending the monsoon poleward, the profile of ascent is roughly constant, with a weak maximum at about 600 hPa, similar to the oceanic profile in Figs. 12a and 12b. In the west, where the low-level flow ventilates the continent, the profile of ascent has, in most models,

two distinct maxima at 850 and 300 hPa, an accentuated version of the land profile in Figs. 12a and 12b. This structure is suggestive of a bimodal distribution of convective motions: either weak and capped at low level by the dry flow or, when CAPE is finally released, deep and intense. These results have a nice correspondence with those of Smyth and Ming (2021), who also find differences in the vertical profile of ascent in idealized simulations of the South American monsoon, depending on the characteristics of the surface. In their case, shallow ascent corresponded to a dry surface and deep ascent to a wet surface. The more realistic land surface, with a bucket model of soil moisture, presented double maxima reminiscent of a mixture of the two soil moisture end members.

6. Summary and discussion

In this paper we have examined the rainband that develops over the idealized tropical continent in the LandControl simulations of the TRACMIP multimodel ensemble (Voigt et al. 2016). The continental rainband moves farthest poleward around summer solstice, 1–2 months preceding its oceanic counterpart. The rainband width, translation speed, and

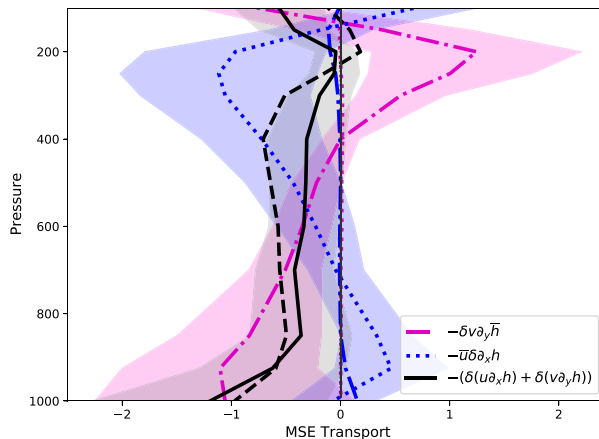


FIG. 9. Vertical average profile of summertime (JJA in the NH and DJF in the SH) LandControl minus AquaControl ensemble-mean anomalous MSE transport [$^{\circ}\text{C m s}^{-1} (^{\circ}\text{latitude/longitude})^{-1}$] in the western subtropical corners of the TRACMIP continent. The total advection anomalies are given in black (the sum of the linearized terms is given in the dashed black line). Blue refers to the advection by the mean wind of the anomalous MSE; magenta refers to the advection by the anomalous wind of the mean MSE. Dotted lines refer to zonal terms and dash-dotted lines to meridional terms. The dominant terms are indicated in the legend and are plotted with a shading corresponding to plus or minus one standard deviation in the multimodel ensemble. The vertical axis is pressure in hPa.

maximum rain rate differ modestly between land and ocean. Previous work (Geen et al. 2019) had suggested that subtropical monsoons abruptly develop an approximately angular momentum-conserving circulation, while those in which maximum rainfall remains within about 10° or 15° of the equator show a weaker, smoothly changing circulation (dubbed an ITCZ-like regime). This distinction motivated our investigation of the mechanisms of ventilation that set the poleward reach of the TRACMIP monsoon. We find that the advection of low MSE into the subtropical land by the low-level anomalous meridional wind acting on the background distribution of moist static energy is the predominant mechanism, while the advection of anomalous MSE by the mean westerlies is secondary. This means that what sets the anomalous circulation sets the position of the rainfall maximum. The opposite is also true: the position of the rainfall maximum modifies the circulation [either directly (Rodwell and Hoskins 1996; Chou et al. 2001) or by bringing the circulation into an approximately angular momentum-conserving regime (Geen et al. 2019)]. Together, these conditions signify a tight coupling between rainfall and circulation, more so in TRACMIP than in previous studies in which the effect of the background circulation was paramount.

These insights are helpful to assess idealized simulations of the tropical rainbands: what is retained and what is lost when a study eliminates a process or an entire component of the climate system? To begin to answer this question we focus on how different idealizations play out in our study and in the broader theoretical literature on the global monsoon and ITCZ.

a. Vertical wind structure

The pivotal studies of Chou et al. (2001) and Chou and Neelin (2003) were carried out with QTCM-1, the first version of the Quasi-Equilibrium Tropical Circulation Model (Neelin and Zeng 2000; Sobel and Neelin 2006). In its original formulation, QTCM simplified the vertical structure of the atmosphere to one with full-troposphere overturning cells and no boundary layer dynamics. By design, therefore, ventilation was the effect of bulk advection of midlatitude oceanic low-MSE air by the column-integrated westerlies in both the basic state and the anomalous circulation [itself a product of the monsoonal rainfall, as in the work of Rodwell and Hoskins (1996)]. While the distribution of land rainfall in TRACMIP is not qualitatively different from that in QTCM, boundary layer advection of low MSE by the anomalous meridional circulation is the key process. The advection by the free troposphere mean westerlies is an active process, but secondary, and mostly counteracted by meridional advection.

One caveat remains necessary: while the TRACMIP GCMs resolve the boundary layer and the ventilation by the low-level flow, they do not reproduce a continentwide shallow meridional circulation similar to the ones that affect the African and Australian monsoons (e.g., Nie et al. 2010). These regions experience dry ascent poleward of the rainband; instead, the TRACMIP continent experience rainfall in all regions of surface ascent. Consistent with the literature on heat lows (Rácz and Smith 1999) and monsoon extent (Chou and Neelin 2003; Smyth and Ming 2021), we attribute this to the fact that the idealized continent has low albedo and only a partial moisture limitation. Nevertheless, the western region of the continent provides an analog for the interaction between the rainband and the shallow circulation and supports the notion that such interaction is significant (Hill et al. 2017; Shekhar and Boos 2017; Zhai and Boos 2017). Besides limiting the extent of the monsoon, the dry northerly flow appears to change the profile of ascent, making it less ocean-like and more consistent with the build up of CAPE and the occurrence of more intense deep convection.

b. Continental geometry

We have not investigated land geometry per se, but we have demonstrated a primary role for the low-level circulation in ventilating the monsoon, and we can speculate on how the continental geometry would matter, at least for equatorial continents. First of all, we can assume that a continent that extended poleward into the region of surface westerlies would be responsive to those as well (just as the TRACMIP ocean responds to the advection by the mean easterlies; Biasutti et al. 2021). Second, because northerly MSE advection is key, a continent that extended into colder oceans, or that included a desert to its poleward flank, would experience greater ventilation. We do not see a straightforward extension of our results to subtropical continents, with ocean on their equatorial boundaries, and thus we can only refer to the relevant literature for such case (see Maroon and Frierson 2016; Maroon et al. 2016; Zhou and Xie 2018; Hui and Bordoni 2021, among others).

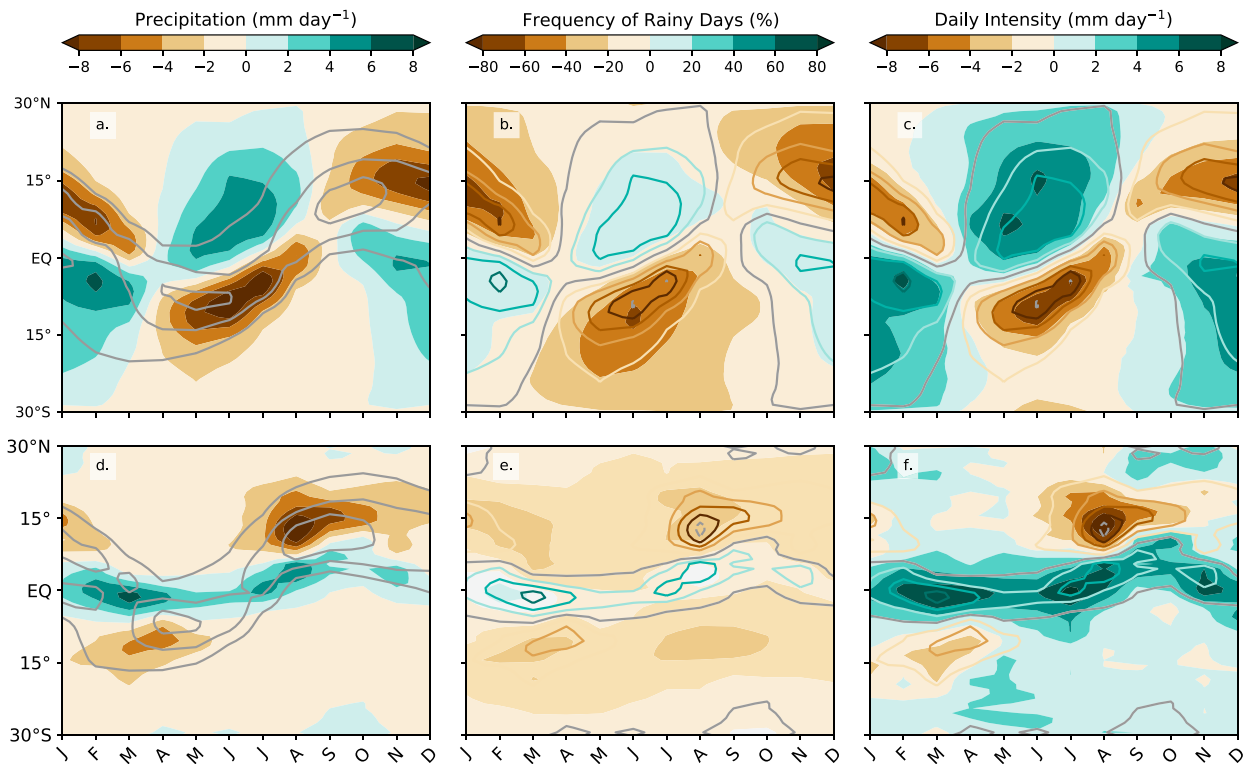


FIG. 10. Latitude–time Hovmöller diagram of climatological LandControl minus AquaControl multimodel-mean monthly anomalies in (a),(d) rainfall (mm day^{-1}), (b),(e) frequency of rainy days (%), and (c),(f) simple daily intensity index (mm day^{-1}). Superimposed on the shaded fields are the AquaControl climatological rainfall (gray contours) in (a) and (d) and the LandControl – AquaControl monthly rainfall anomalies in (b), (c), (e), and (f) [contours colored according to the color bar in (a)]. (top) The mean of the models with reduced heat capacity over land; (bottom) the average of the models with unchanged heat capacity.

We can also speculate on the effect of the width of the continent (again, for the case of a continent straddling the equator). Zhou and Xie (2018) suggested that the length scale of the oceanic influence over land is given by a balance between the time scales of upper-level advection and convective

mixing. But their view presupposes that westerly cold advection is the predominant mechanism of ventilation. If boundary layer processes are instead predominant, the scale of the low-level continental low becomes key. How the latter depends on local and remote rainfall and cloud anomalies, as well as on

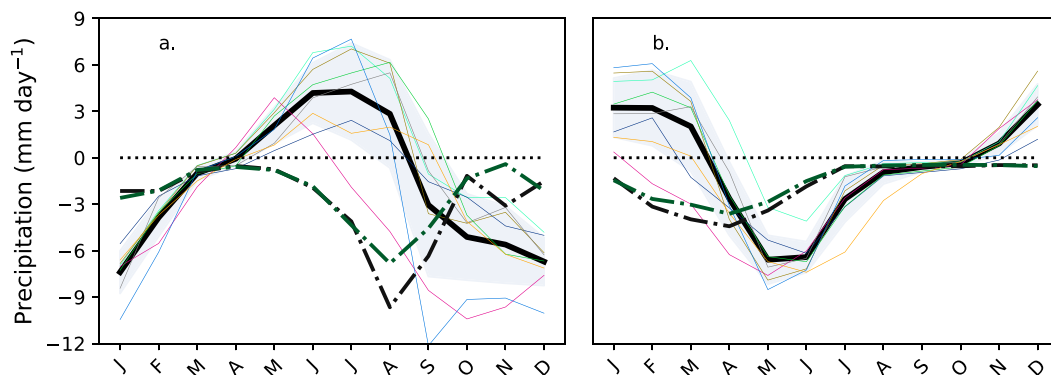


FIG. 11. The annual cycle of LandControl minus AquaControl rainfall anomalies averaged zonally (LandControl data over the continental sector only) and over (a) 10° – 15° N and (b) 10° – 15° S. The solid thin colored lines are individual models that followed protocol, the thick solid black line is their multimodel mean, and the light shading indicates a spread of ± 1 standard deviation. The dash-dotted lines are the two models that did not reduce heat capacity in the continental area.

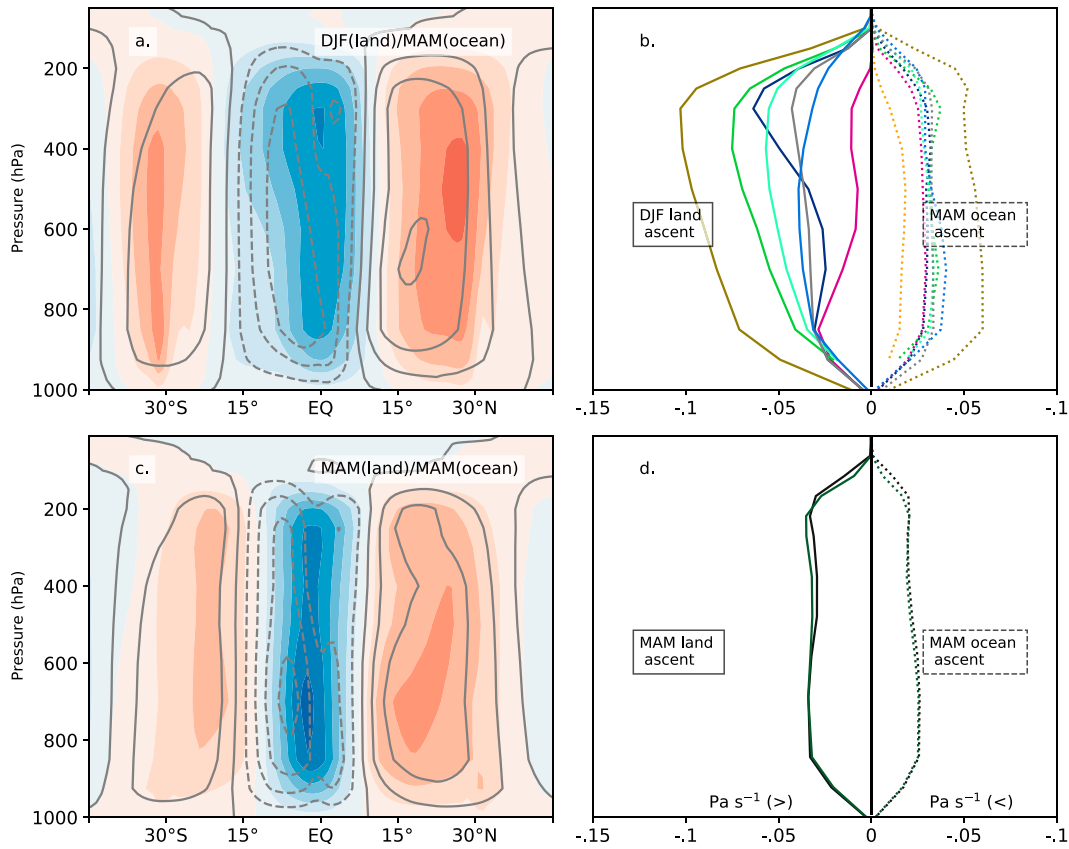


FIG. 12. The effect of heat capacity on ascent profiles. (a) Vertical velocity ω profiles during SH summer in LandControl (DJF, shaded, averaged over the continental sector) and AquaControl (MAM contours, zonally averaged). Fields are the multimodel mean of the models that followed the full TRACMIP protocol in setting up land points. (b) As in (a), but averaged over the latitude of tropical ascent and plotted for each individual model (solid for LandControl and dashed for AquaControl; note that we plot negative values both right and left of the vertical zero line, to allow for a cleaner comparison of the profile shape in LandControl (left) and AquaControl (right)). (c),(d) As in (a) and (b), but for the average of the two MetUM models, which did not reduce land's heat capacity. Southern Hemisphere summer season is therefore defined as MAM for both LandControl and AquaControl.

the characteristics of the lower boundary, remains an open question.

c. Surface evaporation

Previous literature has shown that the treatment of evaporation over land modulates the extent of the monsoon in several key ways, and results from TRACMIP add some detail to this view. Continental rainfall decreases with increased moisture limitation (Chou et al. 2001; Smyth and Ming 2021, although the latter study finds substantial rainfall even for dry land surface) as well as with decreased MSE (Hurley and Boos 2013). We interpret these relationships to mean that impaired evaporation makes continental rainfall more sensitive to ventilation. Based on the similar monsoon limitations in the Chou et al. (2001) and Smyth and Ming (2021) studies (where a bucket model mimicked soil moisture processes) and TRACMIP (where an evaporative resistance crudely mimicked vegetation) we speculate that, as long as evaporation is reduced over land, the means of such reduction might

not be a crucial choice. Second, reduced evaporation contributes to the asymmetry between continental winter cooling and summer warming and to the drying of the equatorial ocean (Biasutti et al. 2021), modulating surface temperature and pressure gradients. Therefore, we expect that the strength and the structure of the low-level circulation anomalies that ventilate the monsoon would depend on the amount of evaporation at the land surface. Third, the absence of a dry shallow circulation in TRACMIP supports the hypothesis that a subtropical desert is necessary for the development of a continent-wide heat low and to the longitudinal extension of the northerly ventilation across the monsoon.

d. Moist radiative processes

Influential studies of the rainbands (ITCZ and monsoons; e.g., Bordoni and Schneider 2008; Bischoff and Schneider 2014; Zhou and Xie 2018, among many) have been carried out with a model (Frierson et al. 2007) that simplified atmospheric physics, and in particular did not include the radiative

effects of water vapor and clouds. Reassuringly, the same model is shown here to behave consistently with the ensemble of protocol models. Nevertheless, the CALTECH model (with no moist-radiative feedbacks; Bordini and Schneider 2008) and the NorESM model (with strong moist-radiative feedbacks; Biasutti et al. 2021) often stand out as outliers. This supports the conclusion of many previous studies [e.g., Kang et al. 2009; Maroon and Frierson 2016; Byrne and Zanna 2020; Biasutti et al. 2021; and for a comprehensive review, see Voigt et al. (2021)] that moist-radiative processes affect the dynamics of the tropical rainbands in important ways, albeit they might not alter their dynamics in a fundamental, qualitative, way.

e. Outlook

The above discussion points to the need to better formalize a modeling hierarchy for land. For other GCM components besides land, there is a recognized hierarchy of model complexity from which researchers can choose the level best suited to their objectives. For example, the ocean can be represented with fixed (uniform or nonuniform) surface temperatures, a slab with specified q fluxes, a column ocean, or a full dynamical ocean (Jeevanjee et al. 2017). It is not obvious what the equivalent hierarchy should look like for land models since there are so many potential properties to include and there might be different combinations with similar complexity. The land geometry is of the utmost importance, but this choice is all but dictated by what monsoon is of interest: it is clear from previous literature that one must at least distinguish between subtropical and tropical monsoons (Geen et al. 2020; Zhou and Xie 2018; Hui and Bordini 2021). As for how to represent the land surface, we agree that heat capacity is the most consequential of the land characteristics, the zeroth order influence on the timing and the strength of the solstitial circulations. Nevertheless, the TRACMIP experiments suggest that anything that affects surface evaporation is also a fundamental knob, capable of shaping the regional circulation and the type of monsoon regime that ensues. Interactive soil moisture [as in Chou et al. (2001)] or vegetation (as, most crudely, in TRACMIP) both fit the bill and so does albedo, which determines the energy available to evaporation, and surface roughness, which alters wind and thus evaporation. The ways in which these factors affect low-level MSE have not been investigated in the theoretical literature as thoroughly as for heat capacity and continental geometry, and they deserve a deeper exploration. The recent work by Smyth and Ming (2021), which investigates both albedo and soil moisture, is a much welcome addition to the canon, but it is mostly limited to a simplified-physics GCM. We suggest the need for more sensitivity experiments with full-physics comprehensive GCM in which the defining factors of land (surface roughness, albedo, soil moisture, and vegetation) can be explicitly tuned for their effect on evaporation. This will allow a land model hierarchy to come into greater focus, and we will be closer to the ideal where anyone who wants to study monsoon dynamics will have a clearly defined array of tools from which to select the one best suited to their research.

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